Maintenance Checklist

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#	Task Name	Task Description	Manual	Page
1	After every run: clean furnace entry disc (remove soot, wipe with IPA)	Remove soot deposits from the entry disc after every run and clean with isopropyl alcohol.	63mm Furnace Manual (Soreq)	9
2	Before operation: check top sleeve + water-cooled top ring for soot; clean if needed	Check graphite top sleeve and water-cooled top ring for soot build-up; remove and clean if necessary.	63mm Furnace Manual (Soreq)	22
3	Before operation: inspect graphite set; clean/replace if needed	Inspect furnace graphite for contamination, wear, or physical damage; clean or replace as required before running.	63mm Furnace Manual (Soreq)	22
4	Before operation: select correct entry sleeve and clean	Select the appropriate entry sleeve size and clean it (as required) before operation.	63mm Furnace Manual (Soreq)	22
5	Before use: if storage vacuum rose >0.03 bar since last use, perform cyclic purge	If the pressure has risen by more than 0.03 bar since previous use, cyclically purge before use.	63mm Furnace Manual (Soreq)	14
6	Cyclic purge after atmosphere exposure (minimum 5 cycles; end in vacuum)	After cool-down ≥20 minutes with water+argon running: vacuum to below -0.85 bar, then argon to ~-0.2 bar; repeat a minimum of 5 cycles; end with vacuum for storage.	63mm Furnace Manual (Soreq)	12
7	Every 3 months OR 1500 operating hours: check pyrometer alignment & window cleanliness	Inspect pyrometer alignment and check window cleanliness; clean window if required.	63mm Furnace Manual (Soreq)	22
8	Every 3 months OR 1500 operating hours: inspect graphite insulation/radiation shield; replace if needed	Inspect graphite insulation/radiation shield for wear or damage; replace as required.	63mm Furnace Manual (Soreq)	22
9	Every 6 months OR 3000 operating hours: check busbar overlap joints tightness	Check overlap joints in the busbar system for tightness.	63mm Furnace Manual (Soreq)	22
10	Every 6 months OR 3000 operating hours: inspect vacuum pump filter cleanliness	Check the cleanliness of the vacuum pump filter; clean/replace as required.	63mm Furnace Manual (Soreq)	22
11	Every 6 months OR 3000 operating hours: tower alignment check using plumb line	Check alignment of all equipment on the tower using a plumb line (see Alignment section).	63mm Furnace Manual (Soreq)	22
12	Monthly: inspect bottom doors for distortion/damage; replace if needed	Inspect furnace bottom doors for distortion or damage and replace if required.	63mm Furnace Manual (Soreq)	22
13	On-condition: inspect/replace bottom sleeve top cap when eroded (e.g., length reduction >3 mm)	Visually inspect bottom sleeve top cap for uneven/lopsided erosion and compare length to new. Replace if overall length reduced significantly (example: >3 mm) or unexplained fibre strength reduction occurs.	63mm Furnace Manual (Soreq)	17
14	On-condition: replace heating element when voltage trend increases (e.g., >2 V above new)	Track voltage/current to maintain temperature. Replace element when increasing voltage trend is established (example criterion: >2 V above new) or if hot/cold spots, fibre tension drift, or unexplained strength reduction indicates expiry.	63mm Furnace Manual (Soreq)	16
15	Weekly: test water and gas flow interlocks	Check the furnace system water and gas flow interlocks (alarms/interlocks section).	63mm Furnace Manual (Soreq)	22
16	Align bare fibre diameter gauge to fibre line using tightened line	Use the tightened line as a fibre surrogate. Move gauge head while observing position on the gauge display until the line is at optical centre.	Fibre Draw Tower Alignment Manual	9

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17	Routine: ensure X/Y slides run free and true	Check that the X/Y slides are running free and true; rectify any stiffness/misalignment.	Bottom Centering Pulley with Break Detector (SG Controls)	4
18	Weekly: check pulley rotates freely, bearing play, and clean V-groove	Check pulley rotates freely and bearing has no significant play. Ensure the root of the pulley 'V' is free from dirt; clean the V with a soft lint-free cloth and methanol or isopropyl alcohol if necessary.	Bottom Centering Pulley with Break Detector (SG Controls)	4
19	Between runs: inspect & clean belts; remove glass debris	Inspect the belts between runs and clean as required. Remove any glass debris from the unit.	Cane / Rod Puller Manual (Soreq)	9
20	Periodic: check carriage movement, belt tension, and lubricate drive linkages	Check for freedom of movement of the carriages and check the tension of all belts. Apply a drop of light machine oil to the drive linkages.	Cane / Rod Puller Manual (Soreq)	9
21	Align cane puller belts to fibre line; record micrometer baseline and offset for cane diameter	Close belts; set right-hand belt (low-pressure) to zero so it can be opened by hand. Set fine micrometer to mid travel. Loosen clamp screws and use pusher blocks to adjust position until line just touches left belt and runs centrally. Retighten. Use micrometer to make final kiss. Record micrometer reading. For canes, offset left belt left by 0.5×cane diameter.	Fibre Draw Tower Alignment Manual	13
22	Set capstan offset (nominal 50 mm) and mark/bolt capstan position	Remove belt and position plumb line near capstan. Set lateral offset ~50 mm (±5 mm) to ensure positive wrap on datum pulley. Align capstan front/back so capstan surface centreline matches the fibre line plane and set pulley parallel to tower face. Mark floor and ideally bolt capstan to the floor.	Fibre Draw Tower Alignment Manual	6
23	As needed: adjust urethane belt tension (cam plate)	Adjust belt tension using cam plate: slacken cam plate clamping screws, pivot cam plate to desired tension, then re-tighten both screws.	Capstan Manual (Capstan V2)	7
24	Before every run: clean pulleys + inspect belt + check surface + confirm belt tension	1) Ensure the capstan assembly and surrounding area are scrupulously clean. 2) Clean the guide pulley and capstan pulley with lint-free tissue soaked in methanol. 3) Inspect the urethane belt for resin debris and embedded fibre; clean or replace if required. 4) Ensure urethane belt tension is correct (minimum tension to prevent slipping). 5) Check capstan surface for scratches/lumps that could damage the fibre.	Capstan Manual (Capstan V2)	6
25	Every 3 months: check bearings are not worn (sealed for life)	Inspect capstan bearings for wear. Bearings are sealed for life; do not lubricate.	Capstan Manual (Capstan V2)	7
26	As needed: replace HEPA filter element (front face access; tighten nuts diagonally)	Isolate power to pre-filter cabinet. From lift/platform, remove securing nuts from retaining clamps while supporting filter. Lift filter forward, install replacement on supporting rail, refit retaining clamps, hand tighten nuts, then fully tighten nuts in diagonal sequence. After filter change, adjust speed control to achieve required HEPA flow.	Clean Air System (SG Controls)	7
27	As needed: replace pre-filter element (with seal inspection and correct airflow direction)	Isolate power. Release cabinet latches, remove outlet to access pre-filter. Remove securing frame and filter element. Inspect PVC foam seals for damage/security. Discard filter, install replacement and securing frame. Ensure filter installed in correct airflow direction (arrows): loose weave faces fan, tight weave faces outlet. Reinstall outlet and close latches; readjust speed control to restore required HEPA flow.	Clean Air System (SG Controls)	6
28	Before drawing: ensure intake air temp matches tower environment; check clean air velocity (0.25–0.5 m/s)	1) Ensure the air drawn into the pre-filter cabinet is at the same temperature as the tower environment to prevent condensation / convection effects. 2) Check airflow from HEPA filters is at a predetermined value within 0.25–0.5 m/s using an anemometer; adjust airflow rate if required (speed control + dampers).	Clean Air System (SG Controls)	3
29	Every 6 months: replace pre-filter; re-adjust speed control; replace HEPA if airflow remains incorrect	1) Replace pre-filter. 2) Reset speed control to give correct HEPA flow rate. 3) If airflow remains incorrect, replace the HEPA filter. Note: after changing any air filter, speed control must be adjusted to achieve required HEPA flow.	Clean Air System (SG Controls)	6

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30	Monthly: inspect wear/damage, verify intakes clear, verify airflow uniform & at optimum; balance dampers	1) Visually inspect system for wear/damage. 2) Ensure air intakes are not obstructed; environment not excessively dusty. 3) Using a hand-held anemometer, verify HEPA airflow at fibre line is at optimum setting and consistent between panels and along panel length. 4) Adjust flow using pre-filter cabinet speed control and manifold dampers to balance flows. 5) If correct flow cannot be obtained via speed control, inspect pre-filter and ensure cabinet intake not blocked.	Clean Air System (SG Controls)	5
31	When switching on: confirm fibre position in bare fibre gauge window stays centered; correct if needed	When the system is operating and the fan comes on line, verify fibre position in the bare fibre diameter gauge does not move too close to the edge of the measurement window. If it does: exit auto diameter control, correct with joystick, and re-check clean air velocity. Also confirm bare fibre tension is correct and fan start line speed is not too low.	Clean Air System (SG Controls)	4
32	Align coated fibre diameter gauge to fibre line using tightened line	Use tightened line as a fibre surrogate and move gauge head until the line appears in the optical centre via the gauge display.	Fibre Draw Tower Alignment Manual	21
33	Align coating resin delivery block to fibre line; record X/Y micrometer readings	Remove coating cartridge assembly, thread tight line through central aperture in resin delivery block. Level block with spirit level and tilt screws. Adjust X/Y using dovetail knobs until centred; insert semi-circular alignment inserts for fine tuning. Record X and Y micrometer positions for quick re-establishment.	Fibre Draw Tower Alignment Manual	16
34	Align concentricity monitor optics and laser aim to fibre line; fine-tune during draw	Loosen clamp screws and nudge baseplate until line is centred in the wrapped screen circle (use vernier). Retighten. Switch on lasers and adjust screws for maximum illumination on the line. Final fine-tune during real draw to maximise fringe clarity.	Fibre Draw Tower Alignment Manual	18
35	Initial setup: establish fibre line using plumb line through furnace to datum pulley	Level the furnace. Fit alignment jigs/irises if available. Move obstructing equipment out of the way. Hang a plumb line from a pin chuck through the furnace. Adjust pin chuck X/Y until line is centred through top iris, then confirm bottom iris closure without line movement (furnace level check). Set bottom datum by moving datum pulley until the line just kisses the root of the V in the datum pulley.	Fibre Draw Tower Alignment Manual	3
36	Periodic verification: check fibre line alignment with tightened line through key stations	Using the established fibre line (tightened line path), verify central passage through furnace, datum pulley, and key process centres (gauges, cooling, capstans/pullers, UV, coating optics). Re-align only if drift is observed.	Fibre Draw Tower Alignment Manual	3
37	3■monthly: grease drum bearing housings	Apply two strokes of grease gun filled with Shell Alvania RA or R2 (or equivalent) to each drum bearing housing; remove surplus grease.	Generic Lubrication Schedule – Fibre Draw Tower (TD0141)	9
38	Monthly: lubricate clamps, guide pulleys, layering arm pulleys	Lightly lubricate drum clamp mechanism, fibre input guide pulleys/spindles, layering arm pulleys/spindles and all pulley bearings; remove excess.	Generic Lubrication Schedule – Fibre Draw Tower (TD0141)	9
39	Daily: cleanliness, smooth operation, pulley condition	Check general cleanliness, smoothness of operation, excessive play, noise or stiffness. Ensure all pulley wheels are clean and free from fibre fragments, dust, or coating residue. Clean using lint-free tissue soaked in methanol if required.	Generic Lubrication Schedule – Fibre Draw Tower (TD0141)	6
40	Monthly: wipe bright shafts and slideways with light oil	Check all bright metal shafts and slideways for cleanliness and smooth operation. Wipe with cloth lightly wetted with light mineral oil (Shell Vactra No.2 or equivalent).	Generic Lubrication Schedule – Fibre Draw Tower (TD0141)	6
41	Monthly (or as required): check tension gauge calibration	Check the tension gauge calibration (tare/span) using the calibration procedure. Include in standard set-up routine if you intend to use the gauge during the draw to correct drift/non-linearity.	Guide Pulley & Tension Gauge (SG Controls)	10
42	On-condition: return load cell to manufacturer if erratic / cannot calibrate	There are no serviceable parts in the load cell. If the unit will not calibrate or gives erratic readings, return it to the manufacturer for attention.	Guide Pulley & Tension Gauge (SG Controls)	10
43	Weekly: check pulley rotation, bearing play, and clean V-groove	Check that the pulley rotates freely, the bearing has no significant play, and that the root of the pulley 'V' is free from dirt. Clean if required.	Guide Pulley & Tension Gauge (SG Controls)	10

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44	Align He cooling tube entry/exit orifices to fibre line; balance ram stops	Close tube around tight line and assess centring at entry/exit. Adjust front/back using clamp screws and front/back adjusting screws. Adjust left/right by moving mechanical set screw stops on the right-hand high-pressure rams; adjust all similarly to keep smooth closure.	Fibre Draw Tower Alignment Manual	10
45	Monthly: lubricate slide bearings on cooling tubes	Lightly lubricate the three pairs of slide bearings fitted to each cooling tube using light mineral oil; remove excess.	Generic Lubrication Schedule – Fibre Draw Tower (TD0141)	7
46	Monthly: lubricate dancer pulley spindles and bearings	Lightly lubricate outrigger and dancer pulley spindles and pulley bearings using light mineral oil; remove excess.	Generic Lubrication Schedule – Fibre Draw Tower (TD0141)	8
47	Align mini capstan pinch roller to fibre line (coarse + fine adjustment)	Close mini capstan on tight line; hold open low-pressure side. Adjust mounting plate (coarse clamp screws) for left/right and forward/back so the line runs down middle of grip surface and just kisses left pulley at its stop. Use fine adjustment screw for final left/right.	Fibre Draw Tower Alignment Manual	11
48	Before use: inspect rollers, clean rollers, check speed control, check local/remote operation	Inspect main and guide rollers for embedded glass and general cleanliness. Clean rollers using lint-free tissue soaked in methanol. Check that motorised roller speed varies correctly when the SET SPEED potentiometer is adjusted. Verify correct operation of the mini-capstan from both local and remote control positions.	Mini Capstan (SG Controls)	8
49	Weekly: inspect cleanliness, roller bearings, carriage bearings, glass embedment, lubricate, check pneumatic leaks	Inspect general cleanliness. Inspect roller bearing wear and carriage roller bearing wear. Inspect for glass embedded in rollers. Clean and lightly lubricate carriage guides and rollers. Check for pneumatic air leaks.	Mini Capstan (SG Controls)	9
50	Weekly: lubricate carriage guides and roller bearings	Clean and lightly lubricate carriage guides and roller bearings with light mineral oil (Shell Vactra No.2 or equivalent); remove excess.	Generic Lubrication Schedule – Fibre Draw Tower (TD0141)	7
51	3-monthly: grease ball screw	Apply grease to the ball screw using the grease nipple with a grease gun.	Preform Feed Assembly (SG Controls)	8
52	3-monthly: grease linear rails	Apply grease to the linear rails using the grease nipple with a grease gun.	Preform Feed Assembly (SG Controls)	8
53	3-monthly: lubricate X-Y slide bearings	Apply a small amount of light machine oil to the X-Y slide bearing parts and exercise the slide mechanism to work the oil in.	Preform Feed Assembly (SG Controls)	8
54	3-monthly: lubricate ball screw support bearings	Apply a small amount of light machine oil to the ball screw support bearings and keep them clean.	Preform Feed Assembly (SG Controls)	8
55	3-monthly: lubricate clamp moving parts	Apply a small amount of light machine oil to the moving parts of the preform clamp and exercise the clamp to distribute the oil evenly.	Preform Feed Assembly (SG Controls)	8
56	Daily: inspect cleanliness and smooth operation of clamp and X-Y slides	Check for general cleanliness and smoothness of operation of the preform clamp assembly and the X-Y centring slides. Remove any debris or contamination.	Preform Feed Assembly (SG Controls)	8
57	3■monthly: grease linear bearings and ball screw (if required)	Apply Shell Alvania RA or R2 (or equivalent) light grease to linear bearing blocks via grease nipples. Lightly smear surplus grease on guide rails. Inspect ball screw and add grease only if required.	Generic Lubrication Schedule – Fibre Draw Tower (TD0141)	6
58	6■monthly: lubricate actuator O■ring seals	Apply a thin film of silicone grease to actuator O■ring seals.	Generic Lubrication Schedule – Fibre Draw Tower (TD0141)	6
59	Monthly: lubricate applicator gimbals and X■Y adjustment mechanism	Lightly lubricate applicator gimbals and the adjustment screw and pivot bearing of the X■Y axis mechanism with light mineral oil; remove excess.	Generic Lubrication Schedule – Fibre Draw Tower (TD0141)	7
60	Before every run/start-up: remove fibre debris; inspect & clean drum surface	1) Remove any fibre debris from the drum carriage, winder frame, and floor area surrounding the winder. 2) Rotate the drum slowly and check the surface for damage, fibre debris, tape, or other contamination. 3) Clean drum surface if required with an alcohol-moistened lint-free wipe.	Take Up Winder (Traversing Drum Type)	6

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61	Every 3 months: grease rotating shaft; wipe & grease guide rails (thin film only)	1) Clean the surface of the rotating shaft with tissue moistened with alcohol, then apply a MoS2-free ball bearing grease (e.g., SKF Alfalub LGMT, Shell Alvania R2, Esso Beacon 2, or BP Energrease LS2) using a cloth, spreading as thinly as possible. 2) Move the drum carriage to one traverse limit and wipe the exposed guide rails; repeat for the opposite end.	Take Up Winder (Traversing Drum Type)	6
62	Every 6 months: inspect drive train (belt tension + contamination/wear/damage)	Remove the drive train cover, check belt tension, and look for signs of contamination, wear, or damage on all components.	Take Up Winder (Traversing Drum Type)	6
63	After full preform draw: inspect quartz liner tube for discolouration (replace if needed)	Inspect the quartz liner tube for any discolouration. Replace the tube if any discolouration is present.	Ultra Violet Curing System – F600S / I606 / P600M	12
64	Align UV lamp entry/exit irises to fibre line (hexapod turnbuckles)	Close lamp entry and exit irises and adjust lamp position until the line is centred in both apertures. Use hexapod turnbuckles to set position.	Fibre Draw Tower Alignment Manual	20
65	Before every start-up: clean shutters and quartz liner tube	Clean shutter doors (remove resin deposits) using lint-free tissue soaked in methanol. Clean quartz liner tube using a long-handled bottle brush soaked in methanol through full tube length several times (top to bottom). Ensure tube is fully seated on locating pin after cleaning.	Ultra Violet Curing System – F600S / I606 / P600M	11
66	Every 500 operating hours: clean lamp module, reflector, bulb and RF screen	Clean lamp module surfaces. Inspect and clean reflector, UV lamp bulb and RF screen using lint-free tissue soaked in methanol.	Ultra Violet Curing System – F600S / I606 / P600M	12
67	Monthly: lubricate UV lamp VAM pivot hinges	Clean and lightly lubricate UV lamp VAM pivot hinges using light mineral oil; remove excess.	Generic Lubrication Schedule – Fibre Draw Tower (TD0141)	8
68	Every 3 months: remove dust from PSU units and replace air filters if required	Remove dust from PSU units. Inspect and replace air filters if required.	Ultra Violet Curing System – F600S / I606 / P600M	12
69	Applicator alignment to tower axis check	Check applicator alignment to tower axis using tensioned fishing line between top/bottom datums. Coarse alignment: center line by eye in adaptor block hole. Fine alignment: use split alignment plate to center line in 1 mm aperture. Record XY micrometer settings.	Wet on Dry Coating System Manual (Soreq)	31
70	Applicator/delivery block levelness check	Use a spirit level to confirm the top face of the delivery/adaptor block is level on both axes. Adjust XY tilt screws as required. Re-check daily or before refitting a cleaned die assembly.	Wet on Dry Coating System Manual (Soreq)	31
71	Delivery block purge before fitting cleaned applicator	With coating cartridge removed, place resin catch bowl under delivery block. Set purge pressure carefully (usually < 1 bar) and open resin valve briefly so resin wells up from exit port; flushes particles and ensures delivery gallery is full. Cover delivery block immediately after purge.	Wet on Dry Coating System Manual (Soreq)	27
72	End-of-draw housekeeping	Verify coating is switched off at end of drawing (with UV lamps, furnace, preform feed, capstan). If not, check fibre break detector. Prevent coating overflow into bubble stripper cap. Remove spilt resin and fibre debris; cover applicator with lint-free wipe; clean mini-capstan rollers.	Wet on Dry Coating System Manual (Soreq)	26
73	Resin level & vessel cleanliness check	Check resin level in all vessels and confirm sufficient resin for intended draw. Remove dust from lids/tops before opening. Check cleanliness/condition of lid gaskets/O-rings; clean or replace as required. Minimize vessel open time; allow time for bubbles to disperse after refilling.	Wet on Dry Coating System Manual (Soreq)	31
74	Resin line purge after refill / dip-tube exposure	After any vessel refill (dip tube removed/raised), air may enter the resin line. Perform a resin line purge to remove trapped air before running (see resin purging section).	Wet on Dry Coating System Manual (Soreq)	32
75	Water circuit checks (weekly) + flush (annual)	Weekly: check water level and condition in the circulator; top up with ready-mixed corrosion inhibitor solution; verify normal water flow and check for leaks. Annual (or sooner if required): drain fully, fill with clean water, circulate several minutes, drain, then refill with inhibitor solution.	Wet on Dry Coating System Manual (Soreq)	33
76	Weekly: joints leakage check + line purge + cleaning	Inspect exposed joints for resin leakage. Perform a line purge on each channel long enough to expel air bubbles. Wipe clean fibre debris, resin spillages, and contamination.	Wet on Dry Coating System Manual (Soreq)	32